Moaaz Emam Ahmed

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EDUCATION

Cairo University ||Faculty of Engineering || GPA: 3.58 2022-2027
Major: Communication and Computer Engineering

SKILLS

- **Programming Languages:** C++, C, C#, Python, JavaScript, SQL, ARM assembly, Verilog HDL, Java
- **Development Tools & Environments:** Git, GitHub, Linux, Visual studio, IntelliJ IDEA, SQL Server, MongoDB, NodeJS, Postman, .NET framework, ModelSim, Logisim, Apache Airflow, Docker, Scrapy, Playwright, n8n, Airtable
- **Soft Skills:** Problem solving, Team collaboration, Rapid adaptability and learning, Effective communication
- **Certifications & Courses:**
    - Backend Development Certificate – IEEE
    - AI Certificate – IEEE

EXPERIENCE

Data Engineering Intern: *Instacodigo (Remote)* Jul 2025 – Oct 2025

- Designed and implemented an automated data pipeline using Apache Airflow for large-scale web data collection, processing, validation, and integration with Odoo ERP.
- Built a modular scraping framework using Scrapy and Playwright to extract structured data from static and JavaScript-rendered websites at scale.
- Implemented data cleaning, deduplication, serialization, and schema enforcement using Pydantic, with branching DAG workflows routing products, jobs, and articles to appropriate Odoo models via OdooRPC.
- Developed an automated social media workflow using n8n and Airtable, supporting scheduled content verification and posting to 13+ platforms with OAuth2 authentication and token refreshing.

PROJECTS

Digital ALU: 2024

- Designed and implemented both software and hardware version of a custom 3 signed-bit ALU supporting 4 modes: addition, subtraction, multiplication, and remainder division.
- Developed the ALU using Verilog HDL and rigorously tested it using ModelSim and testbenches.
- Designed hardware prototype using Logisim and transitioned the design to physically optimized implementation to reduce component count and resource usage to achieve maximum space and cost efficiency.

Automated Garbage Truck with Material Segregation: 2025

- Designed, built, and programmed a fully autonomous garbage-picking car using STM32 Blue pill MCU and ARM Assembly.
- Implemented obstacle avoidance using servo-mounted sensors to scan surroundings, garbage detection and metal-nonmetal segregation using metal detectors, and bin-level detection with an embedded return-to-drop-site mechanism.
- Led the team in hardware implementation and integration, CAD arm design, and optimized sizing and weight distribution, ensuring optimal performance.

OS Scheduler and Memory Manager: 2024

- Developed an OS-level scheduler in C on a Linux environment that selects among 4 implemented scheduling algorithms: SJF, Preemptive HPF, RR (with configurable quantum), Preemptive MLFQ.
- Engineered a process generator to simulate task arrivals and integrated intercross communication using message queues, signals, and shared memory, while employing forking to handle concurrent processes.
- Designed and implemented a Buddy Memory Allocation system using a tree structure with recursive splitting and merging for efficient dynamic memory management and minimized fragmentation.

Custom Programming Language Compiler: 2025

- Designed and implemented a full compiler front-end for a custom programming language using Flex and Bison in C.
- Built lexical analysis, parsing, AST construction, and semantic analysis supporting variables, functions, control flow, loops, scoping, and basic data types.
- Designed grammar rules, symbol table, and three-address code (TAC) intermediate representation.
- Implemented semantic checks including type mismatch, undeclared/redeclared identifiers, and function parameter mismatch, and developed a GUI-based code editor for compilation and visualization.

Alien vs Earth War simulation: 2024

- Developed a simulation for a conflict between two armies using C++. Optimized for large-scale simulations.
- Designed and implemented custom data structures for various army elements to efficiently manage different real-time operations such as insertion, deletion and updates.
- Engineered multiple simulation modes driven by advanced algorithms to compute battle outcomes and generate comprehensive statistics on the battle results.